Number System, Codes, And Conversion

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Introduction to Number Systems

- Number systems are a way to represent numbers in various forms.
- Common number systems include decimal, binary, octal, and hexadecimal.
- Each system has its own base which determines the digits used.
- A number system is defined as a system of writing to express numbers. It is the mathematical notation for representing numbers of a given set by using digits or other symbols in a consistent manner. It provides a unique representation of every number and represents the arithmetic and algebraic structure of the figures. It also allows us to operate arithmetic operations like addition, subtraction, multiplication and division.

Types of Number Systems

There are various types of number systems in mathematics. The four most common number system types are:

1. Decimal number system (Base- 10)
2. Binary number system (Base- 2)
3. Octal number system (Base-8)
4. Hexadecimal number system (Base- 16)

Decimal Number System

- The decimal system is base 10, using digits from 0 to 9.
- It is the most commonly used number system in everyday life.
- Each position in a decimal number represents a power of ten.

The decimal number 1457 consists of the digit 7 in the units position, 5 in the tens place, 4 in the hundreds position, and 1 in the thousands place whose value can be written as:

$(1\times10^3) + (4\times10^2) + (5\times10^1) + (7\times10^0)$

$(1\times1000) + (4\times100) + (5\times10) + (7\times1)$

$1000 + 400 + 50 + 7$

1457

Binary Number System

- The binary system is base 2, using only the digits 0 and 1.
- It is the foundational number system for computer science and digital electronics.
- Each position in a binary number represents a power of two.

2	14	
2	7	0
2	3	1
	1	1

Octal Number System

- The octal system is base 8, utilizing digits from 0 to 7.
- It is often used in computing as a shorthand for binary numbers.
- Each position in an octal number represents a power of eight.

Hexadecimal Number System

- The hexadecimal system is base 16, employing digits from 0 to 9 and letters A to F.
- It is commonly used in programming and digital electronics to simplify binary representation.
- Each position in a hexadecimal number represents a power of sixteen.

Number System Conversion

- Number system conversion involves changing a number from one base to another.
- Common conversions include decimal to binary, binary to decimal, and others.
- Understanding these conversions is essential for working with different number systems.

Decimal to Binary Conversion

- The decimal to binary conversion can be done using the method of successive division.
- For example, to convert the decimal number 10 to binary, divide by 2 and record the remainders.
- The binary equivalent of decimal 10 is 1010.

Binary to Decimal Conversion

- To convert binary to decimal, multiply each binary digit by its corresponding power of two.
- For instance, the binary number 1010 is calculated as $(1\times2^3) + (0\times2^2) + (1\times2^1) + (0\times2^0)$.
- The decimal equivalent of binary 1010 is 10.

Decimal to Octal Conversion

- Decimal to octal conversion can be achieved through successive division by 8.
- For example, dividing the decimal number 65 by 8 gives a quotient of 8 and a remainder of 1.
- The octal equivalent of decimal 65 is 101.

Octal to Decimal Conversion

- To convert octal to decimal, multiply each digit by its corresponding power of eight.
- For instance, the octal number 101 is calculated as $(1\times8^2) + (0\times8^1) + (1\times8^0)$.
- The decimal equivalent of octal 101 is 65.

Decimal to Hexadecimal Conversion

- Decimal to hexadecimal conversion is performed using successive division by 16.
- For example, dividing decimal 255 by 16 gives a quotient of 15 and a remainder of 15 (F in hex).
- The hexadecimal equivalent of decimal 255 is FF.

Hexadecimal to Decimal Conversion

- To convert hexadecimal to decimal, multiply each digit by its corresponding power of sixteen.
- For instance, the hexadecimal number 1A is calculated as $(1 \times 16^1) + (10 \times 16^0)$.
- The decimal equivalent of hexadecimal 1A is 26.

Number System Solved Examples

Example 1:

Convert $(1056)_{16}$ to an octal number.

Solution:

Given, 1056_{16} is a hex number.

First we need to convert the given hexadecimal number into decimal number

$(1056)_{16}$

$= 1 \times 16^3 + 0 \times 16^2 + 5 \times 16^1 + 6 \times 160$

$= 4096 + 0 + 80 + 6$

$= (4182)_{10}$

Now we will convert this decimal number to the required octal number by repetitively dividing by 8.

8	4182	Remainder
8	522	6
8	65	2
8	8	1
8	1	0
	0	1

Therefore, taking the value of the remainder from bottom to top, we get;

$(4182)_{10} = (10126)_8$

Therefore,

$(1056)_{16} = (10126)_8$

Example 2:

Convert $(1001001100)_2$ to a decimal number.

Solution:

$(1001001100)_2$

$= 1 \times 2^9 + 0 \times 2^8 + 0 \times 2^7 + 1 \times 2^6 + 0 \times 2^5 + 0 \times 2^4 + 1 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 0 \times 2^0$

$= 512 + 64 + 8 + 4$

$= (588)_{10}$

Example 3:

Convert 10101_2 into an octal number.

Solution:

Given,

10101_2 is the binary number

We can write the given binary number as,

010 101

Now as we know, in the octal number system,

$010 \rightarrow 2$

$101 \rightarrow 5$

Therefore, the required octal number is $(25)_8$

Example 4:

Convert hexadecimal 2C to decimal number.

Solution:

We need to convert $2C_{16}$ into binary numbers first.

$2C \rightarrow 00101100$

Now convert 00101100_2 into a decimal number.

$101100 = 1 \times 2^5 + 0 \times 2^4 + 1 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 0 \times 2^0$

$= 32 + 8 + 4$

$= 44$

ASCII Code

- ASCII (American Standard Code for Information Interchange) is a character encoding standard.

- It represents text in computers using numeric codes for characters.
- 7-Bit Encoding: ASCII uses 7 bits to represent each character, allowing for 128 possible characters (2^7 = 128).

- Each ASCII character is assigned a unique decimal value ranging from 0 to 127.
- ASCII codes represent text in computers, telecommunications equipment, and other devices. ASCII has just 128 code points, of which only 95 are printable characters, which severely limit its scope.

Extended ASCII and Unicode

- Extended ASCII expands the standard ASCII set to include additional characters.
- Unicode is a more comprehensive character encoding system that supports multiple languages.
- Unicode can represent over a million different characters.

Binary Coded Decimal (BCD)

- BCD is a class of binary encodings for decimal numbers where each digit is represented by its own binary sequence.
- In this code, each decimal digit is represented by a 4-bit binary number.
- BCD is mainly used in digital systems where decimal arithmetic operations are performed like in computers, calculators, display devices, digital sensors, etc.
- For example, the decimal number 93 is represented in BCD as 1001 0011.
- BCD is useful in digital systems where precise decimal representation is required.

Decimal	0	1	2	3	4	5	6	7	8	9
BCD	0000	0001	0010	0011	0100	0101	0110	0111	1000	1001

Gray Code

- Gray code is a binary numeral system where two successive values differ in only one bit.
- It is used in error correction in digital communications and in rotary encoders.
- Gray code minimizes the chance of errors during transitions between values.

Applications of Number Systems

- Different number systems have various applications in computing and electronics.
- Binary is crucial for digital circuits and binary logic systems.
- Hexadecimal is commonly used in programming and memory addressing.

Challenges in Number System Conversion

- Converting between number systems can lead to errors if not done carefully.
- Understanding the base and the method of conversion is crucial for accuracy.
- Practicing conversions can help improve proficiency in handling different number systems.